

The Formation of P–C Bonds Utilizing Organozinc Reagents for the Synthesis of Aryl- and Heteroaryl-Dichlorophosphines

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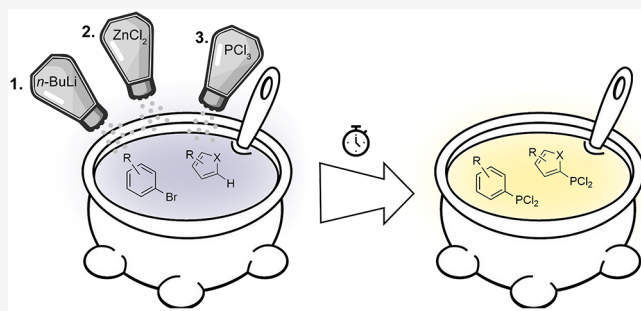


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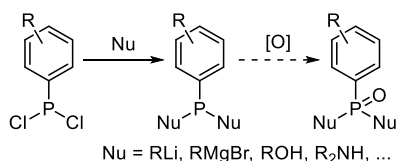
Supporting Information

ABSTRACT: Aryl- and heteroaryl-dichlorophosphines are mildly and selectively made in a one-pot synthesis in moderate to good yields starting from the respective aryl bromides or five-membered heterocycles, following lithiation with *n*BuLi, transmetalation with ZnCl₂, and subsequently the reaction with PCl₃. Selected aryl- and heteroaryl-dichlorophosphines were successfully synthesized using this reaction method and could easily be purified after isolation. The intermediate formation of the organozinc species is essential, as it prevents the formation of multiple substitution products. Important are also the reaction conditions: the usage of the proper solvent for the respective aromatic precursors and removal of the remaining salts by addition of a dioxane/pentane mixture. Depending on the solvent and steric demand of the substituent, mono- and bis-substitution products can be formed but formation also prevented. Hereby, different organozinc species might play an important role.



Dichlorophosphines are precursors in the synthesis of multiple different organic compounds (Scheme 1).^{1–3}

Scheme 1. General Use of Dichlorophosphines in Organic Chemistry



Although dichlorophosphines are widely used in preparative organophosphorus chemistry, the synthesis thereof is still a challenge. The most critical step is the formation of the P–C bond, for which several approaches are known.^{4–7} In most cases, one chlorine atom of PCl₃ is substituted by reaction with a nucleophilic carbon atom (a Grignard reagent or an organolithium compound) yielding the desired dichlorophosphine (Scheme 2, method A).^{8–10} Multiple substitutions, as well as redox reactions, resulting from the nucleophilic attack of the organometallic reagent on chlorine (instead of phosphorus) often occur during these type of reactions. This way, mixtures of multiple substitution products are formed, which are hard to separate from the desired product. In addition, functional groups (ester, C=O in general, CN) are not tolerated, particularly when using organo-lithium reagents.

To avoid multiple substitutions, P(NEt₂)₂Cl is allowed to react with the Grignard reagent or the organolithium compound substituting the sole chlorine atom (Scheme 2,

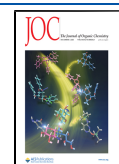
method B). The downside of this method is the step thereafter, where the amino groups are substituted by chlorine when reacting with hydrogen chloride gas.¹¹ Surely, the commercially available HCl dissolved in organic solvents could be used to circumvent the usage of HCl gas. However, dissolved HCl is of low molarity and rather expensive, especially if large quantities of dichlorophosphines (>5 g) want to be prepared.

These difficulties when forming P–C bonds should be overcome by utilizing organozinc compounds. Because of the more covalent character of the C–Zn bond, in contrast to C–Mg and C–Li, the reactivity of the organometallic reagent is significantly reduced, and the selectivity is increased at the same time. These expectations are underlined by many examples for the formation of C–C bonds in organometallic chemistry, e.g., cross-coupling reactions.^{12–16}

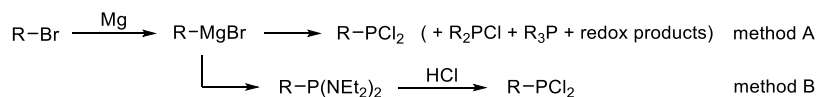
The favored reaction in Scheme 3 should have several advantages, such as the decreased tendency of multiple substitutions and a large toleration of functional groups. In fact, organozinc compounds have been widely used before for the formation of P–C bonds, e.g., in the synthesis of organophosphonates and tertiary phosphines like (cyanometh-

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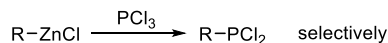
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Scheme 2. Conventional Synthesis of Dichlorophosphines via Methods A and B



Scheme 3. Synthesis of Dichlorophosphines via an Organozinc Reagent

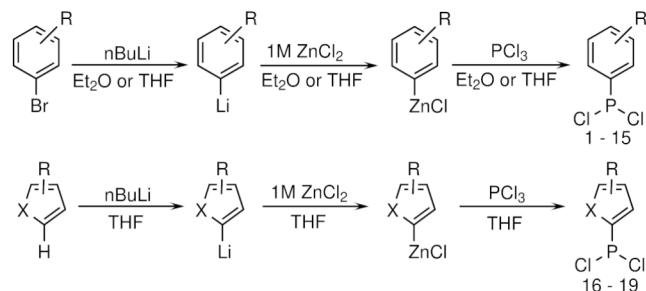


yl)phosphines with two or three cyanomethyl groups, polyfunctional phosphines, or chiral phosphines.^{17–21}

Remarkably, there are only few examples in the literature, which describe the synthesis of dichlorophosphines by reaction of an organozinc reagent with PCl₃. Daugulis et al.²² and Wright et al.²³ used an organozinc reagent for the selective preparation of 4-tolyldichlorophosphine, 4-(*tert*-butyl)phenyl-dichlorophosphine, and 3,5-dimethylphenyldichlorophosphine and utilized these compounds *in situ* for the synthesis of other organophosphorus compounds. Bickelhaupt et al.²⁴ described the synthesis of 2-(dichlorophosphino)-4,5-dimethylphosphinine using the organozinc derivative of the respective phosphinine. Ding et al.²⁵ synthesized the alkyl dichlorophosphine (*n*-oct)PCl₂ starting from the organozinc compound. Last, but not least, Zhu et al.²⁶ prepared 2-pyridyldichlorophosphine utilizing the corresponding organozinc compound. Furthermore, organozinc compounds have been utilized for the synthesis of asymmetric monochlorophosphines in a few publications.^{27–29} Nevertheless, no systematic investigation of the formation of P–C bonds starting from organozinc compounds is found in the literature.

During our systematic investigation on the synthesis of functionalized phosphines, we had a closer look into the formation of P–C bonds with the aid of milder organozinc reagents. Our research is focused on the selective and scalable synthesis as well as the straightforward isolation of clean products in high yields. In that regard, we have examined, as an example, the reaction of aryl- and heteroaryl-Zn compounds with PCl₃ and present herein our results. The emphasis of our investigations is focused on the synthesis of functionalized aryl- or heteroaryl-dichlorophosphines starting from easily accessible aryl bromides or heteroaryl compounds. In the case of dichlorophosphines based on five-membered aromatic rings, the first step corresponds to a simple deprotonation. These dichlorophosphines are of special significance for the synthesis of a variety of organophosphorus compounds.³⁰

The aryl-bromides or heteroaryl compounds are first lithiated with *n*-butyllithium, according to well-known protocols, followed by transmetalation with dry zinc chloride (−78/−100 °C in THF, −10 °C in diethyl ether). THF and diethyl ether are used as solvent for the reactions. During the lithiation, in part cloudy solutions are formed, which mostly clear up after the addition of zinc chloride solution. After reaction of the resulting organozinc species with the equimolar amount of PCl₃ (−78/−100 °C in THF, −10 °C in diethyl ether), the dichlorophosphine is formed under mild conditions and can be isolated after purification in good yields (see Scheme 4 and Table 1). The order of the additions, e.g., if PCl₃ is added to the organozinc compound or the other way around, has no impact on the yields of the reaction products in most cases. Detailed reaction conditions are shown in Table 1 and in the Experimental Section. For almost all reactions, ³¹P NMR

Scheme 4. Synthesis of Aryl- and Heteroaryl-Dichlorophosphines 1–19 via an Organozinc Species^a

^aFor R and X, see Table 1.

spectra measured of samples taken straight from the reaction mixture show a >80% conversion rate of PCl₃ to the desired dichlorophosphine, with the exception being dichlorophosphine 11 (70%).

The main problem, when trying to isolate the dichlorophosphines, is the removal of the inorganic salts ZnCl₂ and LiCl left in the reaction mixture. When carrying out the reaction in THF as a solvent, it is necessary to add pentane and dioxane for the removal of the salts. If only pentane is used, a viscous oil is obtained still containing ZnCl₂ and LiCl. A distillation to isolate and purify the dichlorophosphine is not successful in this case, whereas, when using pentane together with dioxane, after filtration, a subsequent distillation or recrystallization of the products 1–17 is possible. The NMR data of the dichlorophosphines presented here have all been described before in the literature, except for dichlorophosphines 3, 5, 10, 18, and 19, which are listed in the Experimental Section.

When synthesizing the halogen substituted dichlorophosphines 5 and 6, shorter reaction times after the lithiation step are needed to reduce homocoupling reactions. For the synthesis of dichlorophosphines 9, 10, and 16, the respective organozinc species has to be added to PCl₃ in THF to avoid multiple substitutions. When synthesizing dichlorophosphine 11, lower temperatures of −100 °C and below are needed to minimize the formation of side products resulting from nucleophilic additions to the electrophilic carbon atom. The ³¹P NMR spectrum taken from a sample of the reaction mixture of 11 also shows the formation of the respective monochlorophosphine, while supposedly unreacted PCl₃ is also present. The integral ratio of dichlorophosphine to monochlorophosphine to PCl₃ is almost exactly 5:1:1. When the organozinc species is added to PCl₃ in THF, the ³¹P NMR spectrum looks exactly the same. So, reactivity seems not to be the issue at hand. The observations rather indicate a disproportionation of the formed dichlorophosphine 11 in the reaction mixture. Compound 11 on its own has proven to be stable in THF. A complication in the synthesis of 16 is the removal of dioxane during work up and distillation without losing significant amounts of product. A distillation at a higher pressure of 10 mbar was performed, but complete separation could not be achieved. An issue in the synthesis of

Table 1. Reaction Conditions of the Syntheses of the Different Aryl- and Heteroaryl-Dichlorophosphines 1–19^{31,32,34,42,43,a}

R–PCl ₂	Starting material	Reaction conditions	Isolated yield	Yield according to the literature
1		<i>n</i> BuLi/–78 °C/THF, 2 h; ZnCl ₂ in THF (1 M)/–78 °C→RT, 2 h; PCl ₃ /–78 °C→RT, 16 h	63 %	66 % ³¹ (method B)
2		<i>n</i> BuLi/–78 °C/THF, 2 h; ZnCl ₂ in THF (1 M)/–78 °C→RT, 2 h; PCl ₃ /–78 °C→RT, 16 h	55 %	68 % ³¹ (method B)
3		<i>n</i> BuLi/–78 °C/THF, 2 h; ZnCl ₂ in THF (1 M)/–78 °C→RT, 2 h; PCl ₃ /–78 °C→RT, 16 h	82 %	No yields reported ^{22,32}
4		<i>n</i> BuLi/–78 °C/THF, 2 h; ZnCl ₂ in THF (1 M)/–78 °C→RT, 2 h; PCl ₃ /–78 °C→RT, 16 h	87 %	70 % ³³
5		<i>n</i> BuLi/–78 °C/THF, 0.5 h; ZnCl ₂ in THF (1 M)/–78 °C→RT, 2 h; PCl ₃ /–78 °C→RT, 16 h	54 %	60 % ³⁴ (method B)
6		<i>n</i> BuLi/–78 °C/THF, 0.5 h; ZnCl ₂ in THF (1 M)/–78 °C→RT, 2 h; PCl ₃ /–78 °C→RT, 16 h	49 %	26 % ³⁵ (method B)
7		<i>n</i> BuLi/–78 °C/THF, 2 h; ZnCl ₂ in THF (1 M)/–78 °C→RT, 2 h; PCl ₃ /–78 °C→RT, 16 h	65 %	83 % ³⁶ (method A)
8		<i>n</i> BuLi/–78 °C/THF, 1 h; ZnCl ₂ in THF (1 M)/–78 °C→RT, 2 h; PCl ₃ /–78 °C→RT, 16 h	61 %	89 % ³⁶ (method A)
9		<i>n</i> BuLi/–78 °C/THF, 1 h; ZnCl ₂ in THF (1 M)/–78 °C→RT, 2 h; added to PCl ₃ in THF/–78 °C→RT, 16 h	60 %	80 % ³⁷
10		<i>n</i> BuLi/–78 °C/THF, 1 h; ZnCl ₂ in THF (1 M)/–78 °C→RT, 2 h; added to PCl ₃ in THF/–78 °C→RT, 16 h	61 %	New compound
11		<i>n</i> BuLi/–100 °C/THF, 15 min; ZnCl ₂ in THF (1 M)/–100 °C, 30 min; PCl ₃ /–100 °C→RT, 16 h	42 %	44 % ³⁸
12		<i>n</i> BuLi/–78 °C/THF, 2 h; ZnCl ₂ in THF (1 M)/–78 °C→RT, 2 h; PCl ₃ /–78 °C→RT, 16 h	66 %	83 % ⁸ (method B)
13		<i>n</i> BuLi/–78 °C/THF, 2 h; ZnCl ₂ in THF (1 M)/–78 °C→RT, 2 h; PCl ₃ /–78 °C→RT, 16 h	62 %	61 % ³⁹
14		<i>n</i> BuLi/–10 °C/Et ₂ O, 3 h → 30 °C; ZnCl ₂ in Et ₂ O (1 M)/–10 °C → 30 °C, 2 h; PCl ₃ /–10 °C → RT, 16 h	63 %	79 % ⁴⁰ (method B)
15		<i>n</i> BuLi/–10 °C/Et ₂ O, 3 h → 30 °C; ZnCl ₂ in Et ₂ O (1 M)/–10 °C → 30 °C, 2 h; PCl ₃ /–10 °C → RT, 16 h	67 %	76 % ⁴¹ (method A)
16		<i>n</i> BuLi/–78 °C/THF, 2 h; ZnCl ₂ in THF (1 M)/–78 °C→RT, 2 h; added to PCl ₃ in THF/–78 °C→RT, 16 h	39 %	67 % ¹¹ (method B)
17		<i>n</i> BuLi/–78 °C/THF, 2 h; ZnCl ₂ in THF (1 M)/–78 °C→RT, 2 h; PCl ₃ /–78 °C→RT, 16 h	52 %	40–50% ⁴²
18		<i>n</i> BuLi/–78 °C → 0 °C/THF, 1 h; ZnCl ₂ in THF (1 M)/–78 °C→RT, 2 h; PCl ₃ /–78 °C→RT, 16 h	68 %	No yields reported ⁴³
19		<i>n</i> BuLi/–78 °C → RT/THF, 1 h; ZnCl ₂ in THF (1 M)/–78 °C→RT, 2 h; PCl ₃ /–78 °C→RT, 16 h	47 %	Mixture of products ⁴⁴

^aIsolated yields are referenced to the used amount of starting material.

dichlorophosphine **19** is the separation of 2,2'-bithiophene from the final product. A ³¹P NMR spectrum taken from the reaction mixture shows 90% conversion of PCl₃ to the desired dichlorophosphine. However, the ¹H NMR spectrum taken after distillation shows that roughly 20% of 2,2'-bithiophene is present. The most probable cause for this is a disproportionation during distillation, where the respective monochlorophosphine stays in the distilling flask. This was not further investigated.

While for the synthesis of compounds **1–8**, **12**, **13**, and **17–19**, no multiple substitution products were observed in THF, it is of great importance to stick to the right stoichiometry. Already a slight excess of ZnCl₂ in the reaction mixture can trigger the acid-catalyzed competing reaction of THF with PCl₃, as has been described before in the literature.^{45,46} Both degradation products of THF, [Cl(CH₂)₄O]PCl₂ (δ = 178 ppm) and [Cl(CH₂)₄O]₂PCl (δ = 166 ppm), are visible in the ³¹P NMR spectrum of the reaction mixture. For instance, the usage of 1.5 equiv of ZnCl₂ leads to the occurrence of at least 15% of degradation products in total. In the case of the dichlorophosphines **14** and **15**, performing the synthesis in diethyl ether leads to a selective reaction. The products are obtained without any multiple substitution side products or degradation products. The remaining salts can be removed easily by changing the solvent to pentane and purifying the products **14** and **15** by recrystallization.

When attempting the synthesis of dichlorophosphines **1–3** in diethyl ether, surprisingly, the monochlorophosphines are formed almost exclusively, as indicated by the ³¹P NMR spectrum of the reaction mixture. The influence of the solvent on the substitution at phosphorus is apparently dependent on the sterical demand of the organometallic reagent. Bulky substituents in the *ortho*-position to the carbanionic carbon atom lead to the sole synthesis of the dichlorophosphines **14** and **15** in diethyl ether. This unanticipated disparity raised the question of the identity of the organozinc reagents in THF and diethyl ether. Therefore, NMR measurements of the reaction mixtures of the organozinc species resulting from *p*-tolyl–Li (**I**) and (2,4,6-triisopropylphenyl)–Li (**II**) after transmetalation with ZnCl₂ in THF or diethyl ether at RT were performed. The ¹³C NMR spectra clearly show a difference depending on the solvent for species (**I**), while no clear difference can be observed for species (**II**) (see Figure S1 in Supporting Information (SI)). The measured NMR spectra and a more detailed analysis can be found in the SI.

The synthesis of aryl- and heteroaryl-dichlorophosphines starting from the respective aryl-Li/heteroaryl-Li compound via transmetalation onto an organozinc intermediate to lower the reactivity is a convenient alternative to known synthetic procedures of the formation of P–C bonds. This mild reaction can be done in one pot, and the resulting dichlorophosphines can easily be isolated via distillation or recrystallization. For the success of the reaction, it is essential to comply with the right conditions like the solvent used, stoichiometry, and the subsequent workup using dioxane/pentane. While in THF, the dichlorophosphines are obtained simply and successfully, the reaction in diethyl ether is dependent on the bulkiness of the substituent in the *ortho*-position. First NMR spectroscopic investigations of the organozinc species show a divergence in the appearance depending on the aryl-substituent and solvent used, which could be the reason for the different reaction behavior. Future work will be done to further extend this synthetic approach to the synthesis of other dichlorophosphines and to examine the mechanism of their formation in more detail.

EXPERIMENTAL SECTION

All reactions were performed using Schlenk techniques under dry argon. Unless otherwise specified, all reagents and solvents were purchased from commercial sources and used as received. The solvents THF and diethyl ether were dried over elemental sodium and freshly distilled prior to use. CDCl₃ was distilled and stored under an

Ar atmosphere over molecular sieves. NMR spectra were recorded on a Bruker Avance III spectrometer operating at 400.1 MHz (^1H), 161.9 MHz (^{31}P), and 100.6 MHz (^{13}C), as well as on a JEOL Eclipse 270+ instrument operating at 109.4 MHz (^{31}P). Chemical shifts are referred to TMS (^1H , ^{13}C) and 85% H_3PO_4 (^{31}P) as external standards. All NMR spectra were measured, if not mentioned otherwise, at 25 °C. The assignments of the signals for **3**, **5**, **10**, **18**, and **19** are based on 2D (^1H , ^1H -COSY, ^1H , ^{13}C -HSQC, and ^1H , ^{13}C -HMBC) experiments. The multiplicity of each resonance observed in the NMR spectra is reported as s = singlet; d = doublet; t = triplet; q = quartet; m = multiplet.

Synthesis of Aryl-Dichlorophosphines 1–8, 12, and 13 in THF. General Procedure (For Individual Reaction Times, See Table 1). To a solution of the respective aryl-bromide (1.0 equiv, 2 mL for liquids and 2 g/4 g for solids) in anhydrous THF (20 mL) was added the appropriate amount of *n*-butyllithium (1.0 equiv, 2.4 M solution in hexane) dropwise at -78 °C, and the reaction was stirred for another hour. To the resulting suspension was added a solution of anhydrous zinc(II) chloride (1.0 equiv) in anhydrous THF (1 M) at -78 °C. The reaction was stirred (2 h) while warming up to RT to give a colorless/cloudy solution. Phosphorus trichloride (1.0 equiv) was added dropwise at -78 °C, and the resulting mixture was stirred for another 16 h while warming to RT. Afterward, the solvent was removed under reduced pressure. To precipitate the metal salts, 20 mL of pentane and 20 mL of dioxane were added and the mixture stirred for another hour. After removal of the solids via a Schlenk frit, the remaining solution is concentrated *in vacuo*. The left-over liquid is distilled to receive the purified aryl- and heteroaryl-dichlorophosphines **1–8** in moderate to good yields. Dichlorophosphines **12** and **13** are recrystallized from pentane in good yields.

***p*-Tolylidichlorophosphine (1).** 1.96 g, 63%, colorless liquid, bp 57 °C at 1×10^{-3} mbar, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3 , 109.365 MHz] δ = 162.4 ppm (s) (ref 23, ^{31}P NMR [C_6D_6 , 161.98 MHz, +25 °C] δ = 162.8 (s) ppm, for ^1H and ^{13}C NMR, see ref 23).

4-(*tert*-Butyl)phenylidichlorophosphine (2). 1.45 g, 55%, colorless liquid, bp 72 °C at 1×10^{-3} mbar, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3 , 109.365 MHz] δ = 162.4 (s) ppm (ref 23, $^{31}\text{P}\{^1\text{H}\}$ NMR (161.98 MHz, C_6D_6 , +25 °C), δ = 162.8 (s) ppm, for ^1H and ^{13}C NMR, see ref 23).

3,5-Dimethylphenylidichlorophosphine (3). 2.50 g, 82%, colorless liquid, bp 53–57 °C at 1×10^{-3} mbar, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3 , 161.996 MHz] δ = 162.9 ppm (s), ^1H NMR [CDCl_3 , 400.132 MHz] δ = 7.50 (d, 2H, $^3J_{\text{H-P}}$ = 9.9 Hz, $^4J_{\text{H-H}}$ = 0.7 Hz), 7.18 (m, 1H), 2.39 (q, 6H, 2CH₃, $^4J_{\text{H-H}}$ = 0.7 Hz) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR [CDCl_3 , 100.621 MHz] δ = 21.4 (s, 2 CH₃), 127.8 (d, 2CH, $^2J_{\text{C-P}}$ = 32.0 Hz), 134.7 (s, CH), 138.9 (d, Cq, $^3J_{\text{C-P}}$ = 8.5 Hz), 140.3 (d, Cq, $^1J_{\text{C-P}}$ = 51.4 Hz) ppm.

4-Fluorophenylidichlorophosphine (4). 3.12 g, 87%, colorless liquid, bp 62 °C at 1×10^{-3} mbar, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3 , 109.365 MHz] δ = 159.3 (s) ppm (ref 33, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3 , +25 °C], δ = 158.8 ppm, for ^1H and ^{13}C NMR, see ref 47).

4-Chlorophenylidichlorophosphine (5). 1.20 g, 54%, colorless liquid, bp 49 °C at 1×10^{-3} mbar, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3 , 109.365 MHz] δ = 159.5 (s) ppm; ^1H NMR [CDCl_3 , 400.133 MHz] δ = 7.83 (m, 2H), 7.50 (m, 2H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR [CDCl_3 , 100.623 MHz] δ = 139.4 (d, $^4J_{\text{C-P}}$ = 1.4 Hz), 139.0 (d, $^1J_{\text{C-P}}$ = 53.2 Hz), 131.7 (d, $^2J_{\text{C-P}}$ = 32.0 Hz), 129.4 (d, $^3J_{\text{C-P}}$ = 8.2 Hz) ppm; (ref 5, $^{31}\text{P}\{^1\text{H}\}$ NMR δ = 158.3 ppm).

4-Bromophenylidichlorophosphine (6). 1.07 g, 49%, colorless liquid, bp 60 °C at 1.2×10^{-2} mbar, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3 , 109.365 MHz] δ = 159.3 (s) ppm; (ref 35, $^{31}\text{P}\{^1\text{H}\}$ NMR [121 MHz, CDCl_3 , 293 K], δ = 158.6 ppm, for ^1H and ^{13}C NMR, see ref 35).

4-(Trifluoromethyl)phenylidichlorophosphine (7). 2.08 g, 65%, colorless liquid, bp 82 °C at 1×10^{-3} mbar, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3 , 109.365 MHz] δ = 160.0 (s) ppm (ref 23, $^{31}\text{P}\{^1\text{H}\}$ NMR [121 MHz, C_6D_6 , +25 °C], δ = 157 (s) ppm, for ^1H NMR, see ref 48).

4-Methoxyphenylidichlorophosphine (8). 1.36 g, 61%, colorless liquid, bp 62 °C at 3×10^{-3} mbar, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3 , 109.365 MHz] δ = 164.0 (s) ppm (ref 10, $^{31}\text{P}\{^1\text{H}\}$ NMR [161.98 MHz, CDCl_3 , +25 °C], δ = 162.4 (s) ppm, for ^1H and ^{13}C NMR, see ref 10).

1-Naphthalenyldichlorophosphine (12). 2.93 g, 66%, colorless, amorphous solid, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3 , 109.365 MHz] δ = 163.6 (s) ppm (ref 8, $^{31}\text{P}\{^1\text{H}\}$ NMR [CD_2Cl_2 , +25 °C], δ = 164.2 (s) ppm, for ^1H and ^{13}C NMR, see ref 8).

9-Anthracenyldichlorophosphine (13). 2.70 g, 62%, yellow colored, amorphous solid, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3 , 109.365 MHz] δ = 160.0 (s) ppm (ref 39, $^{31}\text{P}\{^1\text{H}\}$ NMR [121 MHz, C_6D_6 , +25 °C], δ = 159.5 (s) ppm, for ^1H and ^{13}C NMR, see ref 49).

Synthesis of Aryl-Dichlorophosphines 9 and 10 in THF. General Procedure. To a solution of the respective aryl-bromide (2 g, 1.0 equiv) in anhydrous THF (20 mL) was added the appropriate amount of *n*-butyllithium (1.0 equiv, 2.4 M solution in hexane) dropwise at -78 °C, and the reaction was stirred for another hour. To the resulting suspension was added a solution of anhydrous zinc(II) chloride (1.0 equiv) in anhydrous THF (1 M) at -78 °C. The reaction was stirred (2 h) while warming up to RT to give a turquoise/cloudy solution. Afterward, the solution containing the organozinc species was added dropwise to phosphorus trichloride (1.0 equiv) in THF (10 mL) at -78 °C, and the resulting mixture was stirred for another 16 h while warming to RT. Afterward, the solvent was removed under reduced pressure. To precipitate the metal salts, 20 mL of pentane and 20 mL of dioxane were added and the mixture was stirred for another hour. After removal of the solids via a Schlenk frit, the remaining solution was concentrated *in vacuo*. The resulting solids were recrystallized from pentane to afford aryl-dichlorophosphines **9** and **10** in good yields.

4-(Dimethylamino)phenylidichlorophosphine (9). 1.33 g, 60%, colorless, amorphous solid, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3 , 109.365 MHz] δ = 165.0 (s) ppm (ref 36, $^{31}\text{P}\{^1\text{H}\}$ NMR [101.3 MHz, C_6D_6 , +25 °C], δ = 165.3 (s) ppm, for ^1H and ^{13}C NMR, see ref 37).

4-(Pyrrolidin-1-yl)phenylidichlorophosphine (10). *N*-(4-Bromophenyl)pyrrolidine was synthesized according to a literature procedure.⁵⁰ 1.34 g, 61%, colorless, amorphous solid, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3 , 161.976 MHz] δ = 166.9 (s) ppm; ^1H NMR [CDCl_3 , 400.133 MHz] δ = 7.76 (m, 2H), 6.61 (m, 2H), 3.37 (m, 4H), 2.06 (m, 4H); $^{13}\text{C}\{^1\text{H}\}$ NMR [CDCl_3 , 100.636 MHz] δ = 150.9 (s, Cq) 132.2 (d, 2CH, $^3J_{\text{C-P}}$ = 35.5 Hz), 124.1 (d, Cq, $^1J_{\text{C-P}}$ = 49.4 Hz), 111.5 (d, 2CH, $^3J_{\text{C-P}}$ = 10.8 Hz), 47.7 (s, 2CH₂), 25.5 (s, 2CH₂) ppm; ^{15}N NMR [via ^1H , ^{15}N -HMBC, CDCl_3] δ = -313.8 ppm. EA could not be conducted due to air sensitivity. In HRMS (EI) measurements, only the hydrolyzed product could be observed: HRMS (EI) m/z [$\text{M} - \text{H}$]⁺ calcd. for $\text{C}_{10}\text{H}_{13}\text{NO}_2\text{P}^+$ 210.0678, found 210.0635.

Synthesis of Aryl-Dichlorophosphine 11 in THF. To a solution of 4-bromobenzonitrile (2.0 g, 11.0 mmol, 1.0 equiv) in anhydrous THF (20 mL) was added *n*-butyllithium (11.0 mmol, 1.0 equiv, 2.4 M solution in hexane) dropwise at -100 °C, and the reaction was stirred for 15 min. To the resulting mixture was added a solution of anhydrous zinc(II) chloride (1.5 g, 11.0 mmol, 1.0 equiv) in anhydrous THF (1 M) at -100 °C. The amber-colored solution was stirred for 30 min. Phosphorus trichloride (1.0 equiv) was added dropwise to the mixture at -100 °C and stirred for another 16 h while warming to RT. Afterward, the solvent was removed under reduced pressure. To precipitate the metal salts, 20 mL of pentane and 20 mL of dioxane were added and the mixture was stirred for another hour. After removal of the solids via a Schlenk frit, the remaining solution was concentrated *in vacuo*. The left-over liquid is distilled to receive the purified aryl-dichlorophosphine **11** in moderate yield.

***p*-Cyanophenylidichlorophosphine (11).** 0.94 g, 42%, colorless liquid, bp 140 °C, 4.5×10^{-2} mbar, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3 , 161.835 MHz] δ = 155.6 (s) ppm (ref 38, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3] δ = 155.4 ppm, for ^1H and ^{13}C NMR, see ref 38).

Synthesis of Aryl-Dichlorophosphines 14 and 15 in Diethyl Ether. General Procedure. To a solution of the respective aryl-bromide (5 mL, 1.0 equiv) in anhydrous diethyl ether (30 mL) was added the appropriate amount of *n*-butyllithium (1.0 equiv, 2.4 M solution in hexane) dropwise at -10 °C, and the reaction was stirred for another 3 h at 30 °C. To the resulting suspension was added a solution of anhydrous zinc(II) chloride (2 equiv) in anhydrous diethyl ether (1 M) at -10 °C. The reaction was stirred further for 2 h at RT to give a colorless/cloudy solution. Phosphorus trichloride (1.1 equiv)

was added dropwise to the mixture at $-10\text{ }^{\circ}\text{C}$ and stirred for another 12 h at RT. The solvent is removed in vacuo, and the left-over solid was extracted two times with dry pentane. For further purification, product **14** can be distilled and **15** can be recrystallized from pentane. The aryl-dichlorophosphines **14** and **15** are received in good yields.

(2,4,6-Trimethylphenyl)dichlorophosphine (14). 4.54 g, 63%, colorless liquid, bp $64\text{ }^{\circ}\text{C}$ at 1×10^{-3} mbar, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3 , 161.996 MHz] $\delta = 168.0$ ppm (s) with 6% impurity of the PBrCl compound at 162.2 ppm (s) (ref 40, $^{31}\text{P}\{^1\text{H}\}$ NMR [CD_2Cl_2] $\delta = 168.8$ (s) ppm, for ^1H , see ref 40).

(2,4,6-Triisopropylphenyl)dichlorophosphine (15). 3.84 g, 67%, colorless, amorphous solid, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3 , 161.996 MHz] $\delta = 166.6$ ppm (s) with 6% impurity of the PBrCl compound at 160.5 ppm (s) (ref 41, ^{31}P NMR [C_6D_6] $\delta = 165.2$ ppm, for ^1H and ^{13}C NMR, see ref 51).

Synthesis of Aryl-Dichlorophosphines 16–19 in THF. General Procedure (For Individual Reaction Times and Conditions, See Table 1). To a solution of the respective heteroarene (2 mL, 1.0 equiv) in anhydrous THF (20 mL) was added the appropriate amount of *n*-butyllithium (1.0 equiv, 2.4 M solution in hexane) dropwise at $-78\text{ }^{\circ}\text{C}$, and the reaction was stirred for another hour. To the resulting suspension was added a solution of anhydrous zinc(II) chloride (1.0 equiv) in anhydrous THF (1 M) at $-78\text{ }^{\circ}\text{C}$. The reaction was stirred (2 h) while warming up to RT to give a colorless/cloudy solution. Phosphorus trichloride (1.0 equiv) was added dropwise to the mixture at $-78\text{ }^{\circ}\text{C}$ and stirred for another 16 h. In the case of **16**, the mixture containing the organozinc species is added dropwise to phosphorus trichloride (1.0 equiv) in anhydrous THF (10 mL) at $-78\text{ }^{\circ}\text{C}$. Afterward, the solvent was removed in vacuum. To precipitate the metal salts, 20 mL of pentane and 20 mL of dioxane were added and the mixture was stirred for another hour. After removal of the solids via a Schlenk frit, the remaining solution is concentrated under vacuum. The left-over liquid is distilled to receive the purified aryl-dichlorophosphines **16–19** in moderate to good yields.

2-Furanyldichlorophosphine (16). 1.82 g, 39%, colorless liquid, bp $42\text{ }^{\circ}\text{C}$ at 10 mbar, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3 , 109.365 MHz] $\delta = 124.0$ (s) ppm (ref 11, $^{31}\text{P}\{^1\text{H}\}$ NMR [CD_2Cl_2] $\delta = 123.7$ (s) ppm, for ^1H and ^{13}C NMR, see ref 11).

2-Thienyldichlorophosphine (17). 2.40 g, 52%, colorless liquid, bp $27\text{ }^{\circ}\text{C}$ at 1×10^{-3} mbar, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3 , 161.976 MHz] $\delta = 146.5$ (s) ppm (ref 11, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3] $\delta = 145.5$ (s) ppm, for ^1H and ^{13}C NMR, see ref 11).

5-Methylthien-2-ylidichlorophosphine (18). 2.81 g, 68%, colorless liquid, bp $45\text{ }^{\circ}\text{C}$ at 1×10^{-3} mbar, ^1H NMR [CDCl_3 , 400.133 MHz] $\delta = 7.47$ (dd, 1H, $^3J_{\text{H-P}} = 7.5$ Hz, $^3J_{\text{H-H}} = 3.6$ Hz), 6.80 (ddq, 1H, $^3J_{\text{H-H}} = 3.9$ Hz, $^4J_{\text{H-P}} = 2.1$ Hz, $^4J_{\text{H-H}} = 1.0$ Hz), 2.59 (t, 3H, $^4J_{\text{H-H}} = 1.1$ Hz); $^{13}\text{C}\{^1\text{H}\}$ NMR [CDCl_3 , 100.636 MHz] $\delta = 152.42$ (d, Cq, $^4J_{\text{C-P}} = 1.2$ Hz), 140.26 (d, Cq, $^1J_{\text{C-P}} = 71.7$ Hz), 136.63 (d, CH, $^2J_{\text{C-P}} = 47.0$ Hz), 126.36 (d, CH, $^3J_{\text{C-P}} = 9.6$ Hz), 15.90 (s, CH_3) ppm; $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3 , 161.976 MHz] $\delta = 145.8$ (s) ppm.

2,2'-Bithiophen-5-ylidichlorophosphine (19). 2,2'-Bithiophene was synthesized according to a literature procedure.⁵² 1.51 g, 47%, colorless liquid, bp $120\text{--}130\text{ }^{\circ}\text{C}$ at 8×10^{-3} mbar, ^1H NMR [CDCl_3 , 400.133 MHz] $\delta = 7.52$ (dd, $^3J_{\text{H-P}} = 7.1$ Hz, $^3J_{\text{H-H}} = 3.8$ Hz, 1H), 7.34 (dd, $^3J_{\text{H-H}} = 5.1$ Hz, $^4J_{\text{H-H}} = 1.2$ Hz, 1H), 7.31 (dd, $^3J_{\text{H-H}} = 3.7$ Hz, $^4J_{\text{H-H}} = 1.2$ Hz, 1H), 7.15 (dd, $^3J_{\text{H-H}} = 3.8$ Hz, $^4J_{\text{H-P}} = 1.8$ Hz, 1H), 7.07 (dd, $^3J_{\text{H-H}} = 5.1$ Hz, $^3J_{\text{H-H}} = 3.7$ Hz, 1H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR [CDCl_3 , 161.976 MHz] $\delta = 148.3$ (d, Cq, $^4J_{\text{C-P}} = 1.2$ Hz), 140.8 (d, Cq, $^1J_{\text{C-P}} = 70.4$ Hz), 136.8 (d, CH, $^2J_{\text{C-P}} = 46.2$ Hz), 135.8 (s, Cq), 128.3 (s, CH), 126.7 (CH), 125.8 (s, CH), 123.7 (d, CH, $^3J_{\text{C-P}} = 9.0$ Hz) ppm; $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3 , 161.976 MHz] $\delta = 144.8$ (s) ppm (ref 44, $^{31}\text{P}\{^1\text{H}\}$ NMR [CDCl_3] $\delta = 144.6$ (s) ppm).

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.joc.1c01560>.

^1H NMR, ^{31}P NMR, $^{13}\text{C}\{^1\text{H}\}$ NMR, heteronuclear NMR, and two-dimensional NMR spectra (PDF)

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Notes

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■ REFERENCES

- (1) Wu, B.; Chopra, R.; Yoshikai, N. One-Pot Benzo[*b*]Phosphole Synthesis through Sequential Alkyne Arylmagnesiation, Electrophilic Trapping, and Intramolecular Phospha-Friedel–Crafts Cyclization. *Org. Lett.* **2015**, *17* (22), 5666–5669.
- (2) Fu, Y.; Hou, G.; Xie, J.; Xing, L.; Wang, L.; Zhou, Q. Synthesis of Monodentate Chiral Spiro Phosphonites and the Electronic Effect of Ligand in Asymmetric Hydrogenation. *J. Org. Chem.* **2004**, *69* (23), 8157–8160.
- (3) Baumgartner, T.; Neumann, T.; Wirges, B. The Dithieno[3,2-*b*:2',3'-*d'*]Phosphole System: A Novel Building Block for Highly Luminescent π -Conjugated Materials. *Angew. Chem., Int. Ed.* **2004**, *43* (45), 6197–6201.
- (4) Buchner, B.; Lockhart, L. B. J. An Improved Method of Synthesis of Aromatic Dichlorophosphines. *J. Am. Chem. Soc.* **1951**, *73*, 755–756.
- (5) Hinke, A.; Kuchen, W. Zur Kenntnis Der Organophosphorverbindungen, XXI Darstellung von Organobromphosphanen Aus RPCl_2 Und R_2PCl Durch Halogenaustausch. *Phosphorus Sulfur Relat. Elem.* **1983**, *15* (1), 93–98.
- (6) Pellon, P.; Hamelin, J. New Access to Functionalized Dichlorophosphines: Synthesis of Two Coordinated Phosphorus Heterocycles. *Tetrahedron Lett.* **1986**, *27* (46), 5611–5614.
- (7) Miles, J. A.; Beeny, M. T.; Ratts, K. W. A General Route to Methoxy-Substituted Arylphosphonous Dichlorides via Mild Lewis Acid Catalysts. *J. Org. Chem.* **1975**, *40* (3), 343–347.
- (8) Reiter, S. A.; Nogai, S. D.; Karaghiosoff, K.; Schmidbaur, H. Insignificance of P–H $\cdots$ P Hydrogen Bonding: Structural Chemistry

of Neutral and Protonated 1,8-Di(Phosphinyl)Naphthalene. *J. Am. Chem. Soc.* **2004**, *126* (48), 15833–15843.

- (9) Karaçar, A.; Freytag, M.; Thönnessen, H.; Jones, P. G.; Bartsch, R.; Schmutzler, R. Preparation and Structures of 1-Dimethylamino-2-Bis(Dimethylamino)- and 1-Chloro-2-Bis(Diethylamino)-1-Phospho-2-Phosphonium Acenaphthene: The First Examples of the 1,2-Dihydro-1,2-Diphospho-Acenaphthene Ring System. *J. Organomet. Chem.* **2002**, *643–644*, 68–80.
- (10) Hettstedt, C.; Unglert, M.; Mayer, R. J.; Frank, A.; Karaghiosoff, K. Methoxyphenyl Substituted Bis(Picolyl)Phosphines and Phosphine Oxides. *Eur. J. Inorg. Chem.* **2016**, *2016* (9), 1405–1414.
- (11) Reiter, S. A.; Nogai, S. D.; Schmidbaur, H. Synthesis and Auration of Primary and Di-Primary Heteroaryl-Phosphines. *Dalton Trans.* **2005**, 247–255, 247.
- (12) Knochel, P.; Molander, G. A. *Comprehensive Organic Synthesis*, 2nd ed.; Knochel, P., Ed.; Elsevier, 2014.
- (13) Knochel, P.; Calaza, M. I.; Hupe, E. Carbon-Carbon Bond-Forming Reactions Mediated by Organozinc Reagents. In *Metal-Catalyzed Cross-Coupling Reactions*; de Meijere, A., Diederich, F., Eds.; Wiley-VCH Verlag GmbH: Weinheim, Germany, 2008; pp 619–670. DOI: 10.1002/9783527619535.ch11.
- (14) Lipshutz, B. H.; Abela, A. R.; Bošković, Ž. V.; Nishikata, T.; Duplais, C.; Krasovskiy, A. Greening up” Cross-Coupling Chemistry. *Top. Catal.* **2010**, *53* (15–18), 985–990.
- (15) Chen, Y. H.; Ellwart, M.; Malakhov, V.; Knochel, P. Solid Organozinc Pivalates: A New Class of Zinc Organometallics with Greatly Enhanced Air- and Moisture-Stability. *Synthesis* **2017**, *49* (15), 3215–3223.
- (16) Enthaler, S.; Wu, X. F. *Zinc Catalysis*; Enthaler, S., Wu, X.-F., Eds.; Wiley-VCH Verlag GmbH & Co. KGaA: Weinheim, Germany, 2015. DOI: 10.1002/9783527675944.
- (17) Verbelen, B.; Dehaen, W.; Binnemans, K. Selective Substitution of POCl₃ with Organometallic Reagents: Synthesis of Phosphinates and Phosphonates. *Synthesis* **2018**, *50* (10), 2019–2026.
- (18) Dahl, O.; Henriksen, U.; Trebbien, C.; Carlberg, G. E.; Sterner, O.; Wickberg, B. Methyl(Cyanomethyl)Phosphines. Synthesis and Nucleophilic Reactivity. *Acta Chem. Scand.* **1983**, *37b*, 639–642.
- (19) Langer, F.; Knochel, P. A New Efficient Preparation of Polyfunctional Phosphines Using Zinc Organometallics. *Tetrahedron Lett.* **1995**, *36* (26), 4591–4594.
- (20) Yakhvarov, D. G.; Ganushevich, Y. S.; Sinyashin, O. G. Direct Formation of P-C and P-H Bonds by Reactions of Organozinc Reagents with White Phosphorus. *Mendeleev Commun.* **2007**, *17* (4), 197–198.
- (21) Langer, F.; Püntener, K.; Stürmer, R.; Knochel, P. Preparation of Polyfunctional Phosphines Using Zinc Organometallics. *Tetrahedron: Asymmetry* **1997**, *8* (5), 715–738.
- (22) Vedejs, E.; Daugulis, O. A Highly Enantioselective Phosphabicyclooctane Catalyst for the Kinetic Resolution of Benzylic Alcohols. *J. Am. Chem. Soc.* **2003**, *125* (14), 4166–4173.
- (23) Solomon, S. A.; Allen, L. K.; Dane, S. B. J.; Wright, D. S. Base-Induced 1,3-Sigmatropic Rearrangement of Mesitylphosphonium Salts. *Eur. J. Inorg. Chem.* **2014**, *2014* (10), 1615–1619.
- (24) Teunissen, H. T.; Bickelhaupt, F. Reactivity of Organozinc Derivatives of Phosphinines. *Organometallics* **1996**, *15* (2), 802–808.
- (25) Wang, Z.; Pu, N.; Tian, Y.; Xu, C.; Wang, F.; Liu, Y.; Zhang, L.; Chen, J.; Ding, S. Highly Selective Separation of Actinides from Lanthanides by Dithiophosphinic Acids: An in-Depth Investigation on Extraction, Complexation, and DFT Calculations. *Inorg. Chem.* **2019**, *58* (9), 5457–5467.
- (26) Chen, X.; Zhu, H.; Wang, T.; Li, C.; Yan, L.; Jiang, M.; Liu, J.; Sun, X.; Jiang, Z.; Ding, Y. The 2V-P,N Polymer Supported Palladium Catalyst for Methoxycarbonylation of Acetylene. *J. Mol. Catal. A: Chem.* **2016**, *414*, 37–46.
- (27) Budzelaar, P. H. M.; Frijns, J. H. G.; Orpen, A. G. Synthesis and Coordination Chemistry of a New Class of Binucleating Ligands: Pyridyl-Substituted Diphosphines. *Organometallics* **1990**, *9* (4), 1222–1227.
- (28) Zhang, H.; Ruiz-Castillo, P.; Buchwald, S. L. Palladium-Catalyzed C–O Cross-Coupling of Primary Alcohols. *Org. Lett.* **2018**, *20* (6), 1580–1583.
- (29) Fukazawa, A.; Usuba, J.; Adler, R. A.; Yamaguchi, S. Synthesis of Seminaphtho-Phospho-Fluorescein Dyes Based on the Consecutive Arylation of Aryldichlorophosphines. *Chem. Commun.* **2017**, *53* (61), 8565–8568.
- (30) Błażewska, K. M.; Drabowicz, J.; Fiaud, J.-C.; Gudat, D.; Hey-Hawkins, E.; Karasik, A. A.; Karpowicz, R.; Kashemirov, B. A.; Kawashima, T.; Kielbasiński, P.; Kilah, N. L.; Kobayashi, J.; Krasowska, D.; Lacour, J.; Lammertsma, K.; Lewkowski, J.; Linder, D.; Łyżwa, P.; Marinetti, A.; McKenna, C. E.; Methot, J. L.; Mikolajczyk, M.; Pietrusiewicz, K. M.; Pirat, J.-L.; Roush, W. R.; Słotweg, J. C.; Stankevič, M.; Stevens, C. V.; Toffano, M.; Urganonkar, S.; Verkade, J. G.; Virieux, D.; Colle, J.-N.; Wild, S. B.; Yoshifuji, M.; Zajac, A. *Category 5, Compounds with One Saturated Carbon-Heteroatom Bond*; Science of Synthesis, Vol. 42; Mathey, F., Trost, B. M., Eds.; Georg Thieme Verlag: Stuttgart, Germany, 2009. DOI: 10.1055/b-003-125759.
- (31) Contrella, N. D.; Sampson, J. R.; Jordan, R. F. Copolymerization of Ethylene and Methyl Acrylate by Cationic Palladium Catalysts That Contain Phosphine-Diethyl Phosphonate Ancillary Ligands. *Organometallics* **2014**, *33* (13), 3546–3555.
- (32) Berthold, D.; Breit, B. Chemo-, Regio-, and Enantioselective Rhodium-Catalyzed Allylation of Triazoles with Internal Alkynes and Terminal Allenes. *Org. Lett.* **2018**, *20* (3), 598–601.
- (33) Grabiak, R. C.; Miles, J. A.; Schwenzer, G. M. Synthesis of Phosphonic Dichlorides and Correlation of Their P-31 Chemical Shifts. *Phosphorus Sulfur Relat. Elem.* **1980**, *9* (2), 197–202.
- (34) Bentrude, W. G.; Fu, J. L.; Rogers, P. E. Free-Radical Chemistry of Organophosphorus Compounds. IV. Polar, Bond Strength, and Resonance Effects on Phosphoranyl Radical Formation. *J. Am. Chem. Soc.* **1973**, *95* (11), 3625–3635.
- (35) den Heeten, R.; Zuidema, E.; Lutz, M.; Spek, A. L.; van Leeuwen, P. W. N. M.; Kamer, P. C. J. Bisxantphos: Stereoselective Synthesis and Coordination Behaviour of a New Class of Cyclic Double Bridged Diphosphines. *J. Organomet. Chem.* **2011**, *696* (19), 3113–3120.
- (36) Coyale, E. E.; Doonan, B. J.; Holohan, A. J.; Walsh, K. A.; Lavigne, F.; Krenske, E. H.; O'Brien, C. J. Catalytic Wittig Reactions of Semi- and Nonstabilized Ylides Enabled by Ylide Tuning. *Angew. Chem., Int. Ed.* **2014**, *53* (47), 12907–12911.
- (37) Jansen, H.; Samuels, M. C.; Couzijn, E. P. A.; Słotweg, J. C.; Ehlers, A. W.; Chen, P.; Lammertsma, K. Reactive Intermediates: A Transient Electrophilic Phosphinidene Caught in the Act. *Chem. - Eur. J.* **2010**, *16* (5), 1454–1458.
- (38) Cui, J.; Marankan, F.; Fu, W.; Crich, D.; Mesecar, A.; Johnson, M. E. An Oxyanion-Hole Selective Serine Protease Inhibitor in Complex with Trypsin. *Bioorg. Med. Chem.* **2002**, *10* (1), 41–46.
- (39) Tsurusaki, A.; Nagahora, N.; Sasamori, T.; Matsuda, K.; Kanemitsu, Y.; Watanabe, Y.; Hosoi, Y.; Furukawa, Y.; Tokitoh, N. Synthesis, Structures, and Reactivity of Kinetically Stabilized Anthryldiphosphene Derivatives. *Bull. Chem. Soc. Jpn.* **2010**, *83* (5), 456–478.
- (40) Freeman, S.; Harger, M. J. P. Influence of Ortho Methyl and Isopropyl Substituents on the Reactivity of N-t-Butyl P-Arylphosphonamidic Chlorides with Isopropylamine and t-Butylamine: Steric Acceleration of Metaphosphonimide Formation by an Elimination Addition Mechanism; Contrasting. *J. Chem. Soc., Perkin Trans. 1* **1987**, 1399–1406.
- (41) Yakhvarov, D. G.; Hey-Hawkins, E.; Kagirow, R. M.; Budnikova, Y. H.; Ganushevich, Y. S.; Sinyashin, O. G. Electrocatalytic Reduction of Aryldichlorophosphines with the (2,2'-Bipyridine)Nickel Complexes. *Russ. Chem. Bull.* **2007**, *56* (5), 935–942.
- (42) Bentov, M.; David, L.; Bergmann, E. D. 916. Organophosphorus Compounds. Part IV. Preparation and Reactions of 2-Thienylphosphonous Dichloride. *J. Chem. Soc.* **1964**, *11* (4750), 4750.
- (43) Han, Z. S.; Wu, H.; Xu, Y.; Zhang, Y.; Qu, B.; Li, Z.; Caldwell, D. R.; Foylark, K. R.; Zhang, L.; Roschhangar, F.; Song, J. J.;

Senanayake, C. H. General and Stereoselective Method for the Synthesis of Sterically Congested and Structurally Diverse P-Stereogenic Secondary Phosphine Oxides. *Org. Lett.* **2017**, *19* (7), 1796–1799.

(44) Freeman, J. L.; Zhao, Q.; Zhang, Y.; Wang, J.; Lawson, C. M.; Gray, G. M. Synthesis, Linear and Nonlinear Optical Properties of Phosphonato-Substituted Bithiophenes Derived from 2,2'-Biphenol. *Dalton Trans* **2013**, *42* (39), 14281.

(45) Albers, T.; Edwards, P. G.; Newman, P. D. Ring-Opening of Tetrahydrofuran with PCl₃ Catalysed by an Ortho-Xylylenedizinc Complex. *Inorg. Chem. Commun.* **2013**, *27*, 163–165.

(46) Peringer, P.; Winkler, P. P. Ring Opening Reactions of Tetrahydrofuran with Phosphorus Halides. *Inorg. Chim. Acta* **1986**, *118* (1), L1–L2.

(47) Pedrini, A.; Bertani, F.; Dalcanale, E. Fluorinated Tetraphosphonate Cavitands. *Molecules* **2018**, *23* (10), 2670.

(48) Clark, T. J.; Rodezno, J. M.; Clendenning, S. B.; Aouba, S.; Brodersen, P. M.; Lough, A. J.; Ruda, H. E.; Manners, I. Rhodium-Catalyzed Dehydrocoupling of Fluorinated Phosphine-Borane Adducts: Synthesis, Characterization, and Properties of Cyclic and Polymeric Phosphinoboranes with Electron-Withdrawing Substituents at Phosphorus. *Chem. - Eur. J.* **2005**, *11* (15), 4526–4534.

(49) Wesemann, J.; Jones, P. G.; Schomburg, D.; Heuer, L.; Schmutzler, R. Phosphorus Derivatives of Anthracene and Their Dimers. *Chem. Ber.* **1992**, *125* (10), 2187–2197.

(50) Noble, A.; MacMillan, D. W. C. Photoredox α -Vinylolation of α -Amino Acids and N-Aryl Amines. *J. Am. Chem. Soc.* **2014**, *136* (33), 11602–11605.

(51) Klöcker, H.; Tendyck, J. C.; Keweloh, L.; Hepp, A.; Uhl, W. 3H-Phosphaallenes Revisited: Facile Synthesis by Hydroalumination of Alkynylphosphines and B-Elimination, Stability and Trapping of Transient Species by Coordination to Transition Metal Atoms. *Chem. - Eur. J.* **2019**, *25* (18), 4793–4807.

(52) Massoumi, B.; Alipour, N.; Fathalipour, S.; Jaymand, M. Nanostructured Poly(2,2'-Bithiophene- Co -3,4-Ethylenedioxythiophene). *High Perform. Polym.* **2015**, *27* (2), 161–170.

■ NOTE ADDED AFTER ASAP PUBLICATION

Scheme 4 was corrected on November 5, 2021.